

Homework 2

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(a) Find $\vec{E}$ at point $P(0, y)$ by evaluating;

$$E(0, y) = \int \frac{\rho_L \cdot dl'}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

Ans: Since $dl' = dx'$

$$\rho_L = \rho_0 |x'| = \begin{cases} \rho_0 x' & , 0 \leq x' \leq L/2 \\ -\rho_0 x' & , -L/2 \leq x' \leq 0 \end{cases}$$

$$(\vec{r} - \vec{r}') = (0, y) - (x', 0) = y\hat{a}_y - x'\hat{a}_x$$

$$|\vec{r} - \vec{r}'| = \sqrt{y^2 + x'^2}$$

$$\text{So, } \vec{E}(0, y) = \int_{x'=-L/2}^0 \frac{-\rho_0 x' (y\hat{a}_y - x'\hat{a}_x)}{4\pi\epsilon_0 (y^2 + x'^2)^{3/2}} \cdot dx' + \int_{x'=0}^{L/2} \frac{\rho_0 x' (y\hat{a}_y - x'\hat{a}_x)}{4\pi\epsilon_0 (y^2 + x'^2)^{3/2}} \cdot dx'$$

$$\vec{E}(0, y) = \frac{\rho_0 \cdot y}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{y^2 + x'^2}} \right]_{x'=-L/2}^0 - \frac{1}{\sqrt{y^2 + x'^2}} \bigg|_{x'=0}^{L/2}$$

$$\vec{E}(0,y) = -\frac{\rho_0 \cdot y}{4\pi\epsilon_0} \cdot \left(\frac{-1}{y} + \frac{1}{\sqrt{L^2/4 + y^2}} \right) + \frac{\rho_0 \cdot y}{4\pi\epsilon_0} \left(\frac{-1}{\sqrt{L^2/4 + y^2}} + \frac{1}{y} \right)$$

$$\vec{E}(0,y) = \frac{\rho_0}{2\pi\epsilon_0} - \frac{\rho_0 \cdot y}{2\pi\epsilon_0 \sqrt{\frac{L^2}{4} + y^2}}$$

* a_x term cancels out due to symmetry property

(b) Find electric potential V by solving; $V(0,y) = \int \frac{\rho_L \cdot dl}{4\pi\epsilon_0 |r-r'|}$

$$\bullet |r-r'| = \sqrt{y^2 + x'^2}$$

$$\bullet dl = dx'$$

$$\bullet \rho_L = \rho_0 |x'| = \begin{cases} \rho_0 \cdot x' & , 0 \leq x' \leq L/2 \\ -\rho_0 \cdot x' & , -L/2 \leq x' \leq 0 \end{cases}$$

$$V(0,y) = \int_{x'=-L/2}^0 \frac{-\rho_0 \cdot x' \cdot dx'}{4\pi\epsilon_0 (y^2 + x'^2)^{1/2}} + \int_{x'=0}^{L/2} \frac{\rho_0 \cdot x' \cdot dx'}{4\pi\epsilon_0 (y^2 + x'^2)^{1/2}}$$

$$= \frac{\rho_0}{4\pi\epsilon_0} \left[-\sqrt{y^2 + x'^2} \Big|_{x'=-L/2}^0 + \sqrt{y^2 + x'^2} \Big|_{x'=0}^{L/2} \right]$$

$$V(0,y) = -\frac{\rho_0 \cdot y}{2\pi\epsilon_0} + \frac{\rho_0}{2\pi\epsilon_0} \sqrt{L^2/4 + y^2}$$

(c) Use result $\vec{E}$ from part (a) to find V by evaluating line integral;

$$V(0, y) = - \int_{\infty}^{P(0, y)} \vec{E} \cdot d\vec{l}$$

to show that the results in (b) and (c) are same.

Ans: $V(0, y) = \int_{y'=\infty}^y E \cdot dl \quad \therefore dl = dy'$

$$= - \int_{y'=\infty}^y \left(\frac{-y' \rho_0}{2\pi\epsilon_0 \sqrt{\frac{L^2}{4} + y'^2}} + \frac{\rho_0}{2\pi\epsilon_0} \right) dy'$$

$$= \frac{\rho_0}{2\pi\epsilon_0} \int_{y'=\infty}^y \frac{y'}{\sqrt{\frac{L^2}{4} + y'^2}} \cdot dy' - \frac{\rho_0}{2\pi\epsilon_0} \int_{y'=\infty}^y dy'$$

$$= \frac{\rho_0}{2\pi\epsilon_0} \cdot \sqrt{\frac{L^2}{4} + y'^2} \Big|_{y'=\infty}^y - \frac{\rho_0}{2\pi\epsilon_0} \cdot y' \Big|_{y'=\infty}^y$$

$V(0, y) = \frac{\rho_0}{2\pi\epsilon_0} \sqrt{\frac{L^2}{4} + y^2} - \frac{\rho_0 \cdot y}{2\pi\epsilon_0}$
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(d) Use the result V from part (b) and use the gradient to obtain E :

$$E(0, y) = -\nabla V$$

and show that the results obtained in (a) and (d) are the same.

$$E(0, y) = - \left(\frac{\partial}{\partial x} \cdot 0 + \frac{\partial}{\partial y} \left(\frac{\rho_0}{2\pi\epsilon_0} \sqrt{\frac{L^2}{4} + y^2} - \frac{\rho_0}{2\pi\epsilon_0} y \right) \right)$$

$$= - \frac{d}{dy} \frac{\rho_0}{2\pi\epsilon_0} \cdot \sqrt{\frac{L^2}{4} + y^2} + \frac{d}{dy} \frac{\rho_0}{2\pi\epsilon_0} y$$

$$= - \frac{\rho_0}{2\pi\epsilon_0} \frac{1}{\sqrt{\frac{L^2}{4} + y^2}} \cdot \frac{d}{dy} \left(\frac{L^2}{4} + y^2 \right) + \frac{\rho_0}{2\pi\epsilon_0}$$

$$= \boxed{- \frac{\rho_0 \cdot y}{2\pi\epsilon_0} \cdot \frac{1}{\sqrt{\frac{L^2}{4} + y^2}} + \frac{\rho_0}{2\pi\epsilon_0}}$$